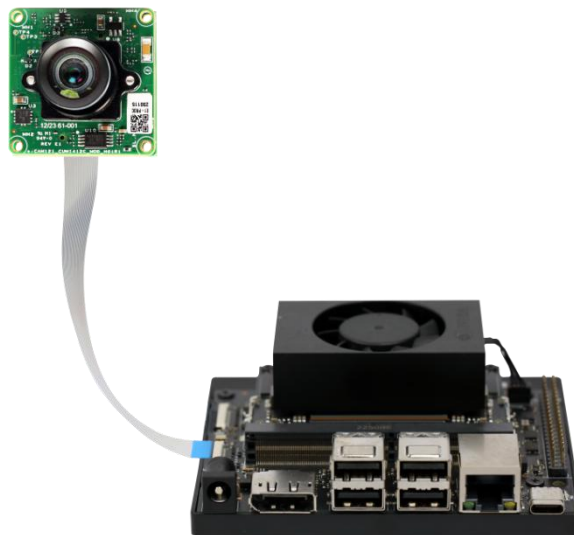


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e-CAM121_CUONX



Datasheet

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1 Introduction

The e-CAM121_CUONX board is a camera board from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. This camera board is directly interfaced with Jetson Orin NX™/Nano™ development kits using J21 connectors.

e-CAM121_CUONX board contains 12.3 MP custom lens camera module based on a IMX412 CMOS image sensor from Sony® STARVIS™. This 12.3 MP color camera has 1/2.3" optical form-factor with electronic rolling shutter and utilizes Jetson™ platforms in-built ISP. It is provided with S-mount lens holder (also known as M12 x P0.5 board lens), which is the most used small form-factor lens mounts for board cameras and offers customized optics.

This document describes about the features of e-CAM121_CUONX board and the pin-outs of the connectors including the mechanical diagram.

2 Disclaimer

The specifications and features of e-CAM121_CUONX camera board are provided here as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

3 Description

e-CAM121_CUONX contains a 12.3 MP custom lens camera module, which is connected to the J21 connectors of the Jetson Orin NX/Nano™ development kits through a 4-Lane Mobile Industry Processor Interface (MIPI) Camera Serial Interface (CSI). Jetson Orin NX™/Nano™ development kit is newly launched small, low-power, AI system-based evaluation kits developed by NVIDIA®.

e-CAM121_CUONX is a multi-board solution, which has two boards as follows:

- Camera module (e-CAM121_CUMI412C_MOD_H01R1)
- Adaptor board (ACC-ONX-MI-WTB-ADP-H01R1)

The front and rear views of the e-CAM121_CUMI412C_MOD_H01R1 board and ACC-ONX-MI-WTB-ADP-H01R1 adaptor board are shown in the below figures.



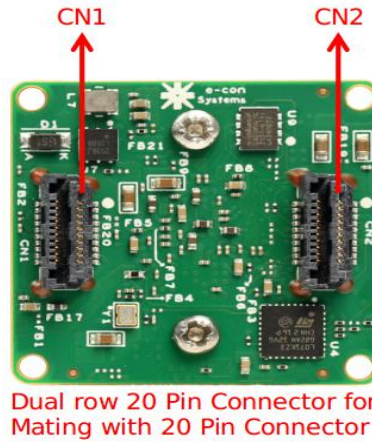


Figure 1: Rear View of e-CAM121_CUMI412C_MOD_H01R1 Module Board



Figure 2: Front View of e-CAM121_CUMI412C_MOD_H01R1 Module Board



Figure 3: Front View of ACC-ONX-MI-WTB-ADP-H01R1 Adaptor Board



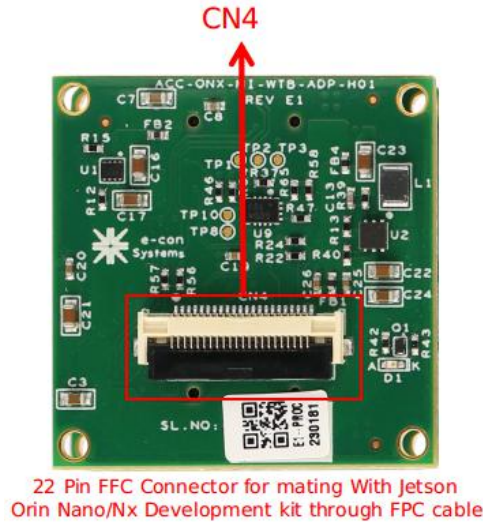


Figure 4: Rear View of ACC-ONX-MI-WTB-ADP-H01R1 Adaptor Board

e-CAM121_CUONX camera module has two 20-pin Samtec connectors (CN1 and CN2) for mating with e-CAM121_CUONX adaptor board. The adaptor board acts as a bridge between the camera module and the NVIDIA® Jetson Orin NX™/Nano™ development kit. It provides the voltages required for the camera module. The e-CAM121_CUONX adaptor board has a 22-pin FFC connector (CN4), through which the camera module is connected to the Jetson Orin NX™/Nano™ development kit using an FPC cable of maximum 15 cm length.

For detailed interfacing of the e-CAM121_CUONX camera board, please refer to the *e-CAM121_CUONX_Getting_Started_Manual_Rev_<ver>.pdf*.

e-CAM121_CUONX operates in two modes as follows:

- [Asynchronous Mode](#)
- [Synchronous Mode](#)

3.1 Asynchronous Mode

The asynchronous mode is the normal streaming mode. In this mode, the camera can be controlled individually.

The following table list the supported frame rates in asynchronous mode.

S. No	Resolution	Frame Rate (fps)
1	4056 x 3040 (10 BPP)	60
2	4056 x 3040 (12 BPP)	40
3	4056 x 3040 (12 BPP)	30



4	4056 x 2288 (10 BPP)	60
5	4056 x 2288 (10 BPP)	30
6	2028 x 1112 (10 BPP)	240
7	2028 x 1112 (10 BPP)	60

Table 1: Resolutions and Frame Rates supported in Asynchronous mode.

3.2 Synchronous Mode

The synchronous mode is the special feature of e-CAM121_CUONX board. e-CAM121_CUONX synchronize all the frames of camera according to the provided PWM pulse as trigger. This PWM trigger pulses can be provided externally (External Trigger Mode).

In External Trigger Mode, e-CAM121_CUONX board has an I/O Header (CN5), through which you can provide PWM pulse of any frequency (less than 60Hz) to synchronize all camera frames. Please ensure to provide trigger signal of 3.3V or 5V for proper functionality. Voltage signal greater than 5V will cause permanent damage to the chip.

3.3 Features

The features of e-CAM121_CUONX are as follows:

- Multi-board solution.
- Compatible with NVIDIA® Orin NX™/Nano™ platforms.
- Standard M12 x P0.5 lens holder for use with customized optics or lenses for various applications.
- Light weight, versatile, and portable design.
- Control for individual cameras and numbers of cameras to be streamed is selectable.
- Imaging applications:
 - 12.3 MP CMOS image sensor with RGB 10-bit/12-bit output format.
 - Field of View (FOV) angle is not the same for all preview resolutions.
- Operating power: 567.6mW (cameras streaming condition).
- Operating temperature: -20C to 60C.
- Restriction of Hazardous Substances (RoHS) compliant.

4 Key Specifications

The below table lists the key specifications of the e-CAM121_CUONX.



Description	Specification
Video Format	NV12
Image Resolution	4056 x 3040 (12.3 MP)
Supported OS	Linux

Table 2: Key Specifications of e-CAM121_CUONX

4.1 CMOS Image Sensor Specification

The below table lists the specification of the CMOS image sensor used in the e-CAM121_CUONX board.

Sensor Specification	
Type/Optical Size	1/2.3" Optical format CMOS image sensor
Resolution	12.3 MP
Sensor Type	RAW 10-bit/12-bit
Pixel Size	1.55 μm x 1.55 μm
Total Number of Pixels	4072H x 3176V
Sensor Effective Area	4072H x 3064V
Active Pixels	4056H x 3040V

Table 3: CMOS Image Sensor Specification

For more information about IMX412 CMOS image sensor or for *Datasheet*, please contact Sony®.

5 Pin Description

The below sections describe the connectors and corresponding pin-outs of the e-CAM121_CUMI412C_MOD_H01R1 module board and the ACC-ONX-MI-WTB-ADP-H01R1 adaptor board.

- [Adaptor Board Connectors](#)
- [Module Board Connectors](#)

5.1 Adaptor Board Connectors

e-CAM121_CUONX adaptor board has three connectors CN1, CN3 and CN4. The CN1 and CN3 are dual row 20-pin Samtec connectors, used for direct mating with the camera module. The CN4 is a single row 22-pin connector, used for connecting with the Jetson Orin NX™/Nano™ development kit through the FPC cable. The dual row connector is 1 to 2 mating type connectors.

The below sections describe the pin descriptions of adaptor board connectors.

- [Pin-out Details of Adaptor Board Dual Row Connectors \(CN1, CN3\)](#)
- [Pin-out Details of Adaptor Board FFC Connector \(CN4\)](#)
- [Pin-out Details of Adaptor Board CN5 connector](#)



5.1.1 Pin-out Details of Adaptor Board Dual Row Connectors (CN1, CN3)

The below table lists the pin-out details of CN1 connector.

CN1 Pin No	Signal Name	Pin Type	Description
1	MIPI_TXDN0	INPUT	MIPI Data Lane 0 Differential Pair -
2	MIPI_TXDN2	INPUT	MIPI Data Lane 2 Differential Pair -
3	MIPI_TXDP0	INPUT	MIPI Data Lane 0 Differential Pair +
4	MIPI_TXDP2	INPUT	MIPI Data Lane 2 Differential Pair +
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog
7	VCC_3P3	POWER	3.3V Power supply for camera board
8	VCC_3P3	POWER	3.3V Power supply for camera board
9	VCC_3P3	POWER	3.3V Power supply for camera board
10	VCC_3P3	POWER	3.3V Power supply for camera board
11	GND	POWER	Ground signal for digital and analog
12	GND	POWER	Ground signal for digital and analog
13	MIPI_TXCN	INPUT	MIPI Clock Lane Differential Pair -
14	MIPI_TXDN3	INPUT	MIPI Data Lane 3 Differential Pair -
15	MIPI_TXCP	INPUT	MIPI Clock Lane Differential Pair +
16	MIPI_TXDP3	INPUT	MIPI Data Lane 3 Differential Pair +
17	VCC_1P8	POWER	1.8V Power supply for camera board
18	GND	POWER	Ground signal for digital and analog
19	VCC_1P8	POWER	1.8V Power supply for camera board
20	VCC_1P8	POWER	1.8V Power supply for camera board

Table 4: Adaptor Board CN1 Connector Pin Description Details

The below table lists the pin-out details of CN3 connector.

CN3 Pin No	Signal Name	Pin Type	Description
1	MIPI_TXDN1	INPUT	MIPI Data Lane 1 Differential Pair -
2	VCC_2P8	POWER	2.8V Power supply for camera board
3	MIPI_TXDP1	INPUT	MIPI Data Lane 1 Differential Pair +
4	VCC_2P8	POWER	2.8V Power supply for camera board
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog
7	SPI_CLK	OUTPUT	Serial Peripheral Interface Clock Signal (TP).



8	CAM_I2C_SCL	OUTPUT	1.8V IO Camera I ² C SCL signal (Internally pulled-up to 1.8V using 4.7K Ω)
9	SPI-SSN	OUTPUT	Serial Peripheral Interface - Slave Select Signal (TP).
10	CAM_I2C_SDA	I/O	1.8V IO Camera I ² C SDA signal (Internally pulled-up to 1.8V using 4.7K Ω)
11	SPI-MOSI	I/O	Serial Peripheral interface - Master Output Slave Input (TP).
12	nCAM_RESET	OUTPUT	1.8V IO camera reset signal (Active low signal)
13	SPI-MISO	I/O	Serial Peripheral Interface - Master Input Slave output (TP).
14	CAM_PWDN	OUTPUT	1.8V IO camera power down signal (Active high signal)
15	GND	POWER	Ground signal for digital and analog
16	RSVD	--	Reserved
17	CAM_TRIGGER	OUTPUT	1.8V IO Trigger signal for camera
18	CAM_ADDRSEL	OUTPUT	Camera Select Signal
19	CAM_FLASH	INPUT	1.8V IO Strobe signal from camera
20	CAM_SHUTTLE	I/O	1.8V IO signal for camera

Table 5: Adaptor Board CN3 Connector Pin-out Details

5.1.2 Pin-out Details of Adaptor Board FFC Connector (CN4)

The below table lists the pin-out details of the CN4 connector.

CN4 Pin No	Signal Name	Pin Type	Description
1	GND	POWER	Ground signal for digital and analog
2	MIPI_TXDN0	OUTPUT	MIPI Data Lane 0 Differential Pair -
3	MIPI_TXDP0	OUTPUT	MIPI Data Lane 0 Differential Pair +
4	GND	POWER	Ground signal for digital and analog
5	MIPI_TXDN1	OUTPUT	MIPI Data Lane 1 Differential Pair -
6	MIPI_TXDP1	OUTPUT	MIPI Data Lane 1 Differential Pair +
7	GND	POWER	Ground signal for digital and analog
8	MIPI_TXCN	OUTPUT	MIPI Clock Lane Differential Pair -
9	MIPI_TXCP	OUTPUT	MIPI Clock Lane Differential Pair +
10	GND	POWER	Ground signal for digital and analog
11	MIPI_TXDN2	OUTPUT	MIPI Lane 2 differential Pair -
12	MIPI_TXDP2	OUTPUT	MIPI Lane 2 differential Pair +
13	GND	POWER	Ground signal for digital and analog
14	MIPI_TXDN3	OUTPUT	MIPI Lane 3 differential Pair -
15	MIPI_TXDP3	OUTPUT	MIPI Lane 3 differential Pair +
16	GND	POWER	Ground signal for digital and analog
17	PWDN_3P3	INPUT	Module board MCU power down signal



18	RSVD	-	Reserved
19	GND	POWER	Ground signal for digital and analog
20	I2C_3P3_SCL	INPUT	3.3V IO I2C SCL signal (Internally pulled-up to 3.3V using 4.7KΩ)
21	I2C_3P3_SDA	I/O	3.3V IO I2C SCL signal (Internally pulled-up to 3.3V using 4.7KΩ)
22	VCC_3P3	POWER	3.3V Power supply for camera board

Table 6: Adaptor Board CN4 Connector Pin Description Details

5.1.3 Pin-out Details of Adaptor Board CN5 connector

The following lists the pin-out details of CN5 connector.

CN5 Pin No	Signal Name	Pin Type	Description
1	VCC_3P3	POWER	3.3V Power supply from module
2	EXT_TRIGGER	INPUT	Camera Trigger signal
3	CAM_SHUTTER	OUTPUT	Shutter signal
4	GND	POWER	Ground signal for digital and analog

Table 7: Adaptor Board CN5 Connector Pin Description Details

5.2 Module Board Connectors

The below table lists the e-CAM121_CUMI412C_MOD has dual row 20-pin SMT connector. The CN1 and CN2 pin numbers, signal names, pin types and their description from sensor perspective.

CN1 Pin No	Signal Name	Pin Type	Description
1	MIPI_DATA2_N	OUTPUT	MIPI Data Lane 2 Differential Pair -
2	MIPI_DATA0_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
3	MIPI_DATA2_P	OUTPUT	MIPI Data Lane 2 Differential Pair +
4	MIPI_DATA0_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog
7	RSVD	-	No Connection
8	VCC_3P3_CONN	POWER	3.3V Power supply for camera boards
9	RSVD	-	No Connection
10	VCC_3P3_CONN	POWER	3.3V Power supply for camera boards
11	GND	POWER	Ground signal for digital and analog
12	GND	POWER	Ground signal for digital and



			analog
13	MIPI_DATA3_N	OUTPUT	MIPI Data Lane 3 Differential Pair -
14	MIPI_CLK_N	OUTPUT	MIPI Clock Lane Differential Pair -
15	MIPI_DATA3_P	OUTPUT	MIPI Data Lane 3 Differential Pair +
16	MIPI_CLK_P	OUTPUT	MIPI Clock Lane Differential Pair +
17	GND	POWER	Ground signal for digital and analog
18	RSVD	-	No Connection
19	RSVD	-	No Connection
20	RSVD	-	No Connection

Table 8: CN1 Pin Descriptions

CN2 Pin No	Signal Name	Pin Type	Description
1	RSVD	-	No Connection
2	MIPI_DATA1_N	OUTPUT	MIPI Data Lane 1 Differential Pair -
3	RSVD	-	No Connection
4	MIPI_DATA1_P	OUTPUT	MIPI Data Lane 1 Differential Pair +
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog
7	CAM_I2C_SCL	INPUT	I2C Clock signal
8	RSVD	-	No Connection
9	CAM_I2C_SDA	I/O	I2C Data signal
10	RSVD	-	No Connection
11	nMCU_RESET	INPUT	Active low reset control signal. Contains on-board RC circuits
12	RSVD	-	No Connection
13	PWDN	INPUT	General purpose Input. Used for firmware update of MCU. For normal operation, drive LOW
14	RSVD	-	No Connection
15	RSVD	-	No Connection
16	GND	POWER	Ground signal for digital and analog
17	RSVD	-	No Connection
18	RSVD	-	No Connection
19	RSVD	-	No Connection
20	RSVD	-	No Connection

Table 9: CN2 Pin Description

5.3 Connector Part Numbers

The below table lists the connectors used in the e-CAM121_CUONX and its compatible mating connectors.



Connector	Description	Manufacturer	Part Number
e-CAM121_CUONX adaptor board dual row connectors (CN1 and CN3) for mating with e-CAM121_CUONX camera module	Board - Board, 20-Pin, 0.635 mm pitch, Vertical SMD connector	Samtec	LSS-110-01-H-DV-A
e-CAM121_CUONX FFC connector (CN4) for connecting with Jetson Orin NX™/Nano™ development kit through FPC cable	FFC Board - Cable, 22-pin, 0.50 mm pitch, Right angle, Flip lock, SMD connector	Molex	5051102291
FPC cable used for connecting e-CAM121_CUONX with Jetson Orin NX™/Nano™ development kit	FPC cable, 22 position, 0.50 mm pitch, 15 CM length, with conductive on top on both side	Molex	151660237

Table 10: Connector Part Numbers

6 Electrical Specification

The electrical specification of the e-CAM121_CUONX are as follows:

- [Functional Temperature Range](#)
- [Recommended Operating Condition](#)

The values described in this section are measured in e-con Systems lab, and this can be used as reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.

6.1 Functional Temperature Range

The below table lists the functional temperature range of the e-CAM121_CUONX.

Temperature Range	Parameter Description
-20C to 60C	Electrically functional operating range

Table 11: Operating Temperature Range

Note: The default lens (optional) supplied with this camera has operating temperature range of -20C to +60C. You can choose wider operating temperature lens as per your requirements.



6.2 Recommended Operating Condition

The below table lists the recommended operating condition of the e-CAM121_CUONX.

Development Kit	Typical Operating Voltage	Typical Power Consumption
With Orin NX™/Nano™	3.3 V	567.6 mW

Table 12: Recommended Operating Condition

7 Mechanical Specifications

e-CAM121_CUONX adaptor board and camera board size are 30 mm x 30 mm. The below sections describe the board drawing and its dimensions.

7.1 e-CAM121_CUONX Adaptor Board Dimension

The front view of the e-CAM121_CUONX adaptor board with mechanical dimensions is shown below.

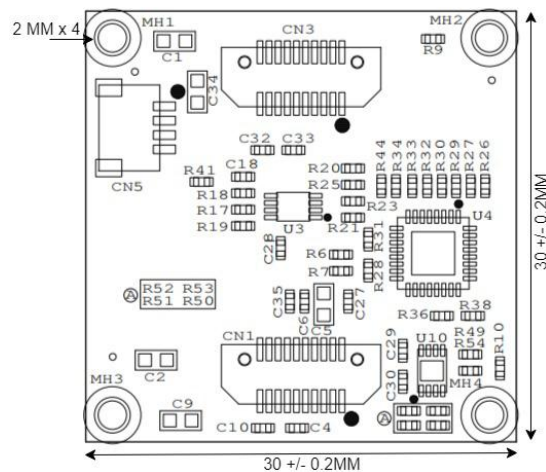


Figure 5: Front View of e-CAM121_CUONX Adaptor Board Mechanical Dimensions

The rear view of the e-CAM121_CUONX adaptor board with mechanical dimensions is shown below.



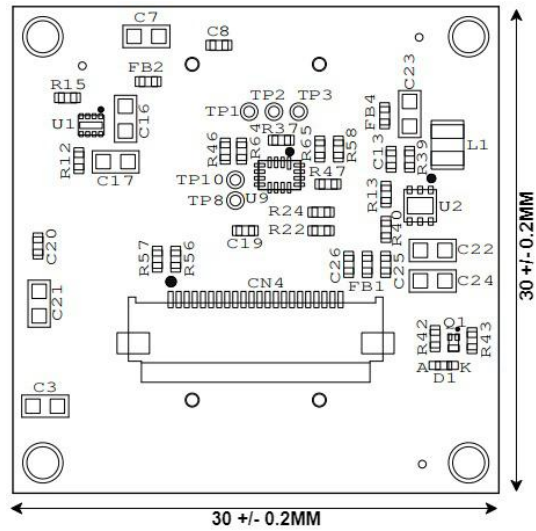


Figure 6: Rear View of e-CAM121_CUONX Adaptor Board Mechanical Dimensions

Note: All dimensions are in mm.

For e-CAM121_CUONX module board mechanical dimension information, please refer to the *e-CAM121_CUMI412C_MOD_Datasheet_Rev_<ver>.pdf*.



8 Revision History

Rev	Date	Description	Author
1.0	03-July-2023	Initial draft	Product Marketing Team
1.1	27-August-2023	Supported Synchronous mode and 4K and 1080p resolution	Product Marketing Team



Support

Contact Us

If you need any support on e-CAM121_CUONX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

